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Key Points:

- The previous upper Paleozoic Gadikao Formation yields high precision Zircon U-Pb dates from ~254 to 248 Ma
- The volcanic rocks yield a robust Permo-Triassic boundary paleopole of 59.7°N and 228.2°E (A_{95} = 3.3°)
- The closure of the Paleo-Tethys Ocean around Qamdo is best estimated at ~232–220 Ma, mostly likely at ~230 Ma

Supporting Information:

- Supporting Information S1
- Table S1
- Table S2
- Table S3
- Table S4
- Table S5
- Table S6

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Paleomagnetic and Chronologic Data Bearing on the Permian/Triassic Boundary Position of Qamdo in the Eastern Qiantang Terrane: Implications for the Closure of the Paleo-Tethys

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Abstract The tectonic evolution of the eastern Paleo-Tethys in the Three-Rivers headwater region remains uncertain, especially because of the lack of quality paleomagnetic data from the east Eastern Qiangtang Terrane (EQT). Here, we present an integrated paleomagnetic and zircon U-Pb geochronologic study of the Gadikao Formation (Fm.) volcanic rocks in the EQT. This previously assumed Carboniferous or Permian rock unit is now zircon U-Pb dated to ~254–248 Ma. Our new high quality (29 sites and 257 samples) inferred primary paleomagnetic results yield a robust Permo-Triassic boundary paleopole of 59.7°N and 228.2°E (A_{95} = 3.3°), with a paleolatitude of 10.0 ± 3.3°N for the study area. Integrated with other reliable Permo-Triassic paleomagnetic data of the EQT and the Tarim Block and other lines of geologic evidence, we propose that the EQT drifted continuously northward during 300–200 Ma and the Paleo-Tethys Ocean probably closed at ~232–220 Ma around the Qamdo region, most likely at 230 Ma.

Plain Language Summary The drift history of the Qiangtang Terrane is the key to understand the evolution of the "Proto-Tibet" and the Paleo-/Meso-Tethys Oceans, which is still under debate. In this work, we provide U-Pb zircon data to constrain (254–248 Ma) a high quality paleomagnetic pole (59.7°N, and 228.2°E, with A_{95} = 3.3°) for the EQT, which was located at ~10.0°N during ~254–248 Ma. Integrated with other available lines of evidence, we propose that the EQT drifted continuously northward during the Permian and Triassic and collided with the Songpan-Ganzi/Tarim Terrane, resulting in the closure of the Paleo-Tethys Ocean around the Qamdo region during ~232–220 Ma, most likely at about 230 Ma.

1. Introduction

The Three-Rivers headwater region (TRHR), located in the eastern Tibetan Plateau, is a complex collage composed of multiple accreted terranes (e.g., Songpan-Ganzi, Eastern Qiangtang, Western Qiangtang, Yidun, and South China) bounded by several sutures, which mark the closure of the eastern Paleo-Tethys Ocean (Figure 1). Many studies have suggested that the drift histories and amalgamation processes of these terranes involved continental-continental collisions (e.g., Chang et al., 1986; Ding et al., 2013; Ma et al., 2019; Metcalfe, 2013; M. D. Yan et al., 2016; X. F. Yang et al., 2017; Y. G. Yan et al., 2019; Yin & Harrison, 2000), accompanied by the evolution of the eastern segment of the Tethys Oceans (e.g., Cao et al., 2019; Dan et al., 2018; Sengör, 1987; M. D. Yan et al., 2016). Thus, recognizing the paleopositions of the above-mentioned terranes and the closure times of the Tethys Oceans are crucial to understand the pre-Cenozoic evolution of the Tibetan Plateau. Over the past decades, although it has generally been accepted that the TRHR was characterized by an archipelago of arcs or continental ribbons separating multiple seaways within the late Paleozoic-early Mesozoic Paleo-Tethys, the evolution process and closure time of the eastern part of the Paleo-Tethys Ocean have remained uncertain (e.g., Reid et al., 2005; Zi et al., 2012).

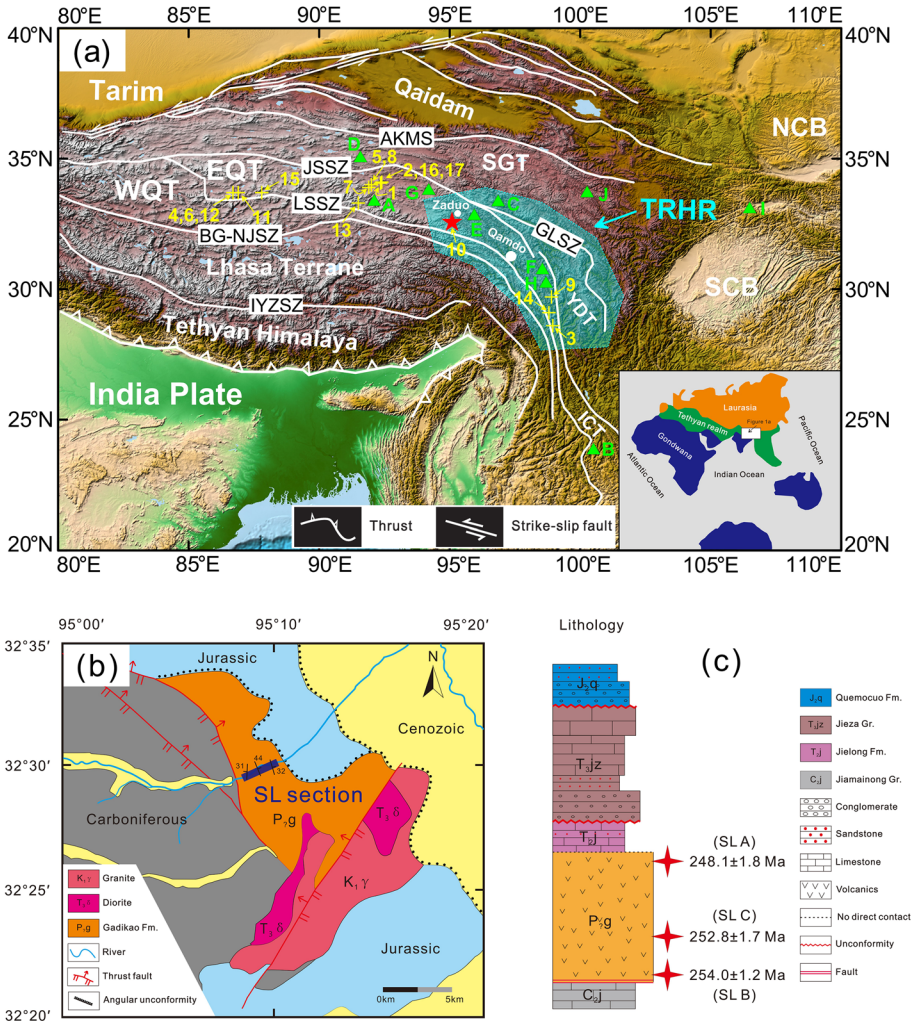


Figure 1. (a) DEM of the Tibetan Plateau and surrounding areas with terranes and major faults, with the inset map showing the Tethyan realm. The red star marks the location of the study area, illustrated in (b). (b) Detailed geologic map of the Sulu section (SL). (c) Lithology and contact relationship of late Paleozoic-early Mesozoic strata in the Zaduo area (modified from QGS, 2005) in the Eastern Qiangtang Terrane. TRHR, Three-Rivers headwater region; AKMSZ, Ayimaqing-Kunlun-MuztaghSuture Zone; JSSZ, Jinsha Suture Zone; GLSZ, Ganzi-Litang Suture Zone; BG-NJSZ, Bangong-Nujiang Suture Zone; LSSZ, Longmu Tso-Shuanghu Suture Zone; IYZSZ, Indus-Yarlung Zangbo Suture Zone; NCB, North China Block; SCB, South China Block; SGT, Songpan-Ganzi Terrane; EQT, Eastern Qiangtang Terrane; WQT, Western Qiangtang Terrane; YDT, Yidun Terrane; ICT, Indochina Terrane. The yellow crosses show the locations of previous studies: 1-X. F. Yang et al., 2017; 2-P. P. Song et al., 2017; 3-X. Z. Li et al., 1999; 4-Cheng et al., 2012; 5-Cheng et al., 2013; 6-Cheng et al., 2012; 7-Ma et al., 2019; 8-Cheng et al., 2013; 9-K. N. Huang et al., 1992; 10-This study; 11-Zhou et al., 2019; 12-Cheng et al., 2012; 13-P. P. Song et al., 2020; 14-X. Z. Li et al., 1999; 15-C. Y. Song et al., 2012; 16-Zhou et al., 2019; 17-P. P. Song et al., 2015 (Table S3). The green triangles represent the geologic evidences supporting the closure of the Paleo-Tethys at early Late Triassic (Table S5). The red four-pointed stars represent the geochronologic sampling positions in the Gadikao Fm.

For example, the estimated closure time of the huge Paleo-Jinsha Ocean ranges from the Late Permian (Sha & Fürsich, 1999) to the Early-Middle Triassic (Pullen et al., 2008; Zi et al., 2012), the Late Triassic (Dan et al., 2018; C. Li et al., 2009; P. P. Song et al., 2015), the Early Jurassic (Ding et al., 2013), the Middle Jurassic (Xiao et al., 2005) or even to the mid-Cretaceous (Yin & Harrison, 2000), based on different approaches from paleontologic, paleomagnetic, magmatic, and sedimentologic analyses.

The Eastern Qiangtang Terrane (EQT) is juxtaposed with the Songpan-Ganzi Terrane, the Western Qiangtang Terrane (WQT), the South China Block (SCB) and the Yidun Terrane along the three major Paleo-Tethyan suture zones, namely, the Jinsha, Longmucuo-Shuanghu and Ganzi-Litang suture zones (Figure 1).

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The EQT is one of the key terranes involved in the tectonic evolution of the Paleo-Tethys (e.g., Gehrels et al., 2011; Metcalfe, 2013; Yin & Harrison, 2000). Determining its drift history is crucial to solve the tectonic evolution of the east Paleo-Tethys.

Paleomagnetism is a powerful method for quantitative paleogeographic reconstruction. Some paleomagnetic studies have been carried out to constrain the late Paleozoic-early Mesozoic formation of the Qiangtang Terrane, such as in the Tuotuohe (Cheng et al., 2013; Song et al., 2015, 2017; X. F. Yang et al., 2017) and Yanshiping areas (M. D. Yan et al., 2016; Ma et al., 2019; P. P. Song et al., 2020) in the central EQT and the Raggyorcaka-Woruoshan area in the west EQT (Cheng et al., 2012; C. Y. Song et al., 2012; Zhou et al., 2019) and WQT (C. Y. Song et al., 2012) (Figure 1a). These results provide a basic framework for the paleolatitude evolution of the Permian-Triassic in this region. To date, however, few late Paleozoic-early Mesozoic paleomagnetic results have been obtained from the eastern EQT, which is a vital linkage region in the TRHR. At present, there is only one Late Permian result (age still controversial) from sedimentary rocks near Markam (K. N. Huang et al., 1992) and two low-quality Permian-Triassic results from Derchin (X. Z. Li et al., 1999; Figure 1). Given the narrow and long (~1,500 km) shape of the EQT, the lack of robust paleomagnetic data from the region have severely hindered our understanding on the paleogeography and the evolution of the Eastern Paleo-Tethys in the TRHR.

In this study, we report results of a new paleomagnetic and U-Pb geochronologic study of the poorly age constrained (previously assigned to be the late Paleozoic) Gadikao Fm. from the eastern EQT, which is now dated to have formed close to the P/T boundary. Integrating our new data with previously available suitable quality paleomagnetic results and other lines of geologic evidence, we refine the Permo-Triassic drift history of the EQT and estimate the closure time of the Paleo-Tethys Ocean in the TRHR.

2. Geologic Setting and Paleomagnetic Sampling

The EQT, also known as the "Northern Qiangtang" Terrane, is located within the central Tibetan Plateau (Figure 1a). Geologically, it is separated from the Songpan-Ganze terrane by the Jinsha suture zone to the north and from the WQT by the Longmu Co-Shuanghu suture zone-Taniantaweng orogenic arc to the south (K. N. Huang et al., 1992; C. Li et al., 2009; M. D. Yan et al., 2016; T. N. Yang et al., 2011). Based on the stratigraphic characteristics, to date, there are no direct records indicating the existence of a metamorphic crystalline basement in the EQT (Pullen et al., 2008). The oldest sedimentary rocks are Early Ordovician in age, covered by Late Devonian to Triassic to Cenozoic sedimentary sequences (Z. Z. Zhao et al., 2001).

The sedimentary sequences in the Zaduo area are the Lower Carboniferous Zaduo Group (Gr.), the Upper Carboniferous Jiamainong Gr., the Permian Kaixinling Gr. and Gadikao Fm., the Middle Triassic Jielong Fm., the Upper Triassic Jieza Gr., and the Middle Jurassic Quemocuo Fm. Two angular unconformities are present between the Middle Triassic Jielong Fm. and the Upper Triassic Jieza Gr. and between the Upper Triassic Jieza Gr. and the Middle Jurassic Quemocuo Fm. (QGS, 2005; Figure 1b and 1c). The rock types of the Gadikao Fm. vary spatially, especially with regard to the abundance of volcanic rocks, forming a volcanic belt along the southern half of the EQT, from Zaduo in the west to Nangqian in the east (QGS, 2005). The Gadikao Fm. is widely exposed in the Zaduo area. It consists mainly layers of rhyolitic tuff, rhyolite dacite, crystalline glassy tuff, and volcanic breccia that are intercalated with limestone and siltstone, with clear paleo-horizontal indicators in the field (Figure S1). Numerous fossils (fusulinids and brachiopods) are present in the limestone beds (QGS, 2005; S. P. Li et al., 2008). The identified fossil brachiopods (e.g., *Gigantoproductus* cf. *giganteus [Sowerby]* and *Delepinea depressa Ching et Liao*) and corals (*Kueichouphyllum* sp.) in the underlying sedimentary unit suggested an Early Carboniferous age (BGE & DQP, 1988), and therefore the Gadikao Fm. was assigned to be an Early Carboniferous rock unit in the 1:200,000 Jiduo regional geologic survey map. Later this formation provided a preliminary U-Pb zircon date of 287 ± 4 Ma (S. P. Li et al., 2008). Because no sampling location, measuring procedure and data details were provided in S. P. Li et al. (2008), the reliability of this data is still questionable. In addition, T. N. Yang et al. (2011) suggested that the volcanic rock dominated Nuoribagaribao and Gadikao Fms. might have formed during the Late Permian. Therefore, major uncertainties still exist for the age of the volcanic rocks in the Zaduo area.

The Sulu (SL) section (32°30.0′N and 95°10.0′E) is exposed as part of an anticline with a SE-NW-trending fold axis (Figure 1b), which was formed during Middle to Late Triassic deformation (QGS, 2005). The folded

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volcanic sequence of the Gadikao Fm., is exposed in two limbs, with clear bedding orientations (Figure 1b and Figure S1). It is in fault contact with the underlying Lower Carboniferous Zaduo Gr. and unconformably overlain by the Middle Jurassic Quemocuo Fm. (Figure 1b). The Gadikao Fm. has a thickness of over 200 m in the SL section. The upper part mainly consists of ~15–20-m-thick at least six rhyolite dacite flows, intercalated with thin-bedded sandstone, which provide both excellent structural control on the original attitude of the lavas and evidence for episodic eruptions over a protracted period of time (Figure S1). Three sites (SL1-3) were collected from the rhyolite dacite flows. Each site contains at least one discrete flow. Underneath the rhyolite dacite flows is a ~5-m-thick volcanic breccia layer, followed by cycles of crystalline glassy tuff and tuff layers (~2–5 m each), where an anticline is present at the bottom of the section. Another 28 sites have been collected along the tuff layers (tuffaceous material fall deposits). A total of 318 core samples from 31 sites were collected from the SL section. In addition, three samples were taken from the stratigraphic top, bottom and middle parts of the Gadikao Fm. for U-Pb zircon age determination. The three samples are the rhyolite dacite (SL A) and the crystalline glassy tuff (SL B) at the northeast flank, and the tuff (SL C) at the southwest side of the anticline, respectively. (Figure S1).

3. Zircon U-Pb Geochronology

Detailed analytical procedure and the cathode luminescence (CL) images for selected samples are described in Text S2 in the supporting information. The cathodoluminescence images show that all the zircon grains from samples SL A, B and C are generally transparent, euhedral to subhedral, have a prismatic shape with clear oscillatory zoning (Figure S2). The lengths of the long axes of zircon grains are 80–150 μm and the Th/U ratios are larger than 0.3, indicative of a magmatic origin (Table S1 in the supporting information). The analyses of 23 zircon grains from the rhyolite dacite sample (SL A) yield a weighted mean $^{206}Pb/^{238}U$ age of 248.1 ± 1.8 Ma (2σ; MSWD = 1.8; Figure S2d); 24 spot analyses of zircons from the crystalline glassy tuff sample (SL B) yield a weighted mean $^{206}Pb/^{238}U$ age of 254.0 ± 1.2 Ma (2σ; MSWD = 0.86; Figure S2e), whereas 23 zircon spots from the tuff sample (SL C), which is stratigraphically between samples SL A and B, yield a weighted mean $^{206}Pb/^{238}U$ age of 252.8 ± 1.7 Ma (2σ; MSWD = 1.6; Figure S2f). Therefore, our detailed U-Pb zircon analyses of the three volcanic samples from the stratigraphic top, middle and bottom part of the Gadikao Fm. (Figures 1c and Figure S1) indicate that they are of volcanic origin and have age estimates of 248.1 ± 1.8 Ma, 252.8 ± 1.7 Ma and 254.0 ± 1.2 Ma, respectively (Figure 1c, Figure S1 and S2). The ages are reliable and consistent with the stratigraphic sequence and show that the age of the Gadikao Fm. was around the P/T boundary. Similar U-Pb zircon data of the Nuoribagaribao Fm. in the Dongmozhazhua area ~40 km east of the Zaduo County were also reported (T. N. Yang et al., 2011), suggesting synchronous volcanic eruptions across a large area of the region.

4. Paleomagnetic Results

Rock magnetic experiments and analysis, and paleomagnetic treatment, including thermal and alternating field demagnetization analysis are described in Texts S3 and S4 in the supporting information. A total of 257 out of 318 specimens yield reliable ChRM directions that provide 29 out of 31 estimated site-mean directions (Table S2). A total of 29 independent sites, 257 specimens (Figure 2, Table S2), with means of 22 normal and 7 reverse polarity. They provide an overall mean direction of $D_g = 8.7°$, $I_g = 36.1°$, $k_g = 16.5$, $\alpha_{95} = 6.8°$ in situ, and $D_s = 22.0°$, $I_s = 19.4°$, $k_s = 38.7$, and $\alpha_{95} = 4.4°$ after tilt correction. The tilt-corrected α_{95} (k) is smaller (larger) than that of in situ (from 6.8° [16.5] to 4.4° [38.7]), indicating an improved clustering of site mean directions after tilt correction. The 29 site means yield positive McFadden (1990) and Watson and Enkin (1993) fold and McFadden & McElhinny (1990) reversals tests (see details in Text S5), as well as pass the paleosecular variation check (Text S5), likely yielding a primary paleopole of 59.7°N and 228.2°E ($A_{95} = 3.3°$), with a paleolatitude of 10.0° ± 3.3°N for the study area.

5. Discussion

5.1. The Northward Drift of the EQT in the Permo-Triassic

In this study, we interpret the obtained ChRM directions to represent a Late Permian to Early Triassic geomagnetic field, and the new paleomagnetic pole (59.7°N, 228.2°E, $A_{95} = 3.3°$) is based on a reasonably large

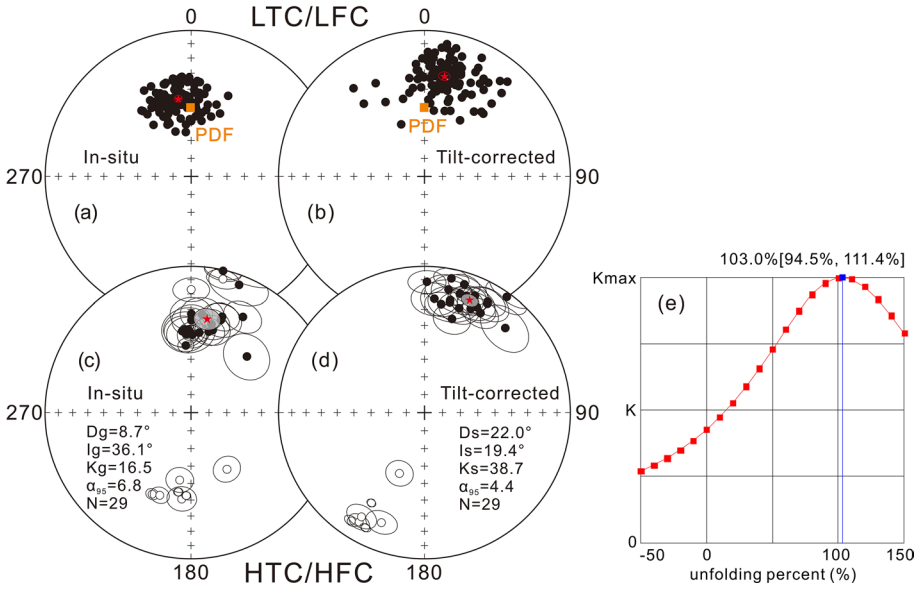


Figure 2. Equal-area projections of the viscous overprinted (low-temperature (field) component) directions (a) and (b) and characteristic remnant magnetization directions (c) and (d) for the Upper Permian- Lower Triassic Gadikao Fm., and the progressive unfolding analysis of the ChRM directions using the Watson and Enkin (1993) method (e). PDF: direction of the present-day geomagnetic field. Open/filled circles represent upward/downward inclinations, respectively.

data set and a high quality (29 sites, 257 samples, fulfill all the Van der Voo's, 1990 seven quality criteria) for the EQT, with a corresponding paleolatitude of ~10.0° ± 3.3°N during the ~254–248 Ma time interval (mean of ~251 Ma) at reference locality 32.5°N and 95.2°E.

To better understand the drift history of the EQT during the Permian to Triassic, all 17 available paleomagnetic results from rocks in the region have been compiled (Table S3; Cheng et al., 2012, 2013, K. N. Huang et al., 1992; Ma et al., 2019; X. Z. Li et al., 1999; Song et al., 2012; Song et al., 2015, 2017, 2020; X. F. Yang et al., 2017; Zhou et al., 2019). Given that ssthe EQT is long (over ~1,500 km) and narrow, all the paleomagnetic directions were converted to the present study site (32.5°N and 95.2°E), the new paleolatitudes were then recalculated and are shown in Figure 3 and Table S3. Because of differences in data quality, we re-evaluate the above paleomagnetic data following the restrictive criteria for high-quality paleomagnetic data that must satisfy the first five criteria of Van der Voo's (1990) (Besse & Courtillot, 2002): (1) well-determined rock age (uncertainties<15 Myr or by accurate isotopic dating); (2) sufficient number of samples ($n \geq 36$ [specimens] and $N \geq 6$ [sites]; k [or K] ≥ 10 and α_{95} [A_{95}] $\leq 15°$); (3) stepwise demagnetization; (4) robust field tests (fold test or reversals test); and (5) structural control and craton coherence.

Among these 17 paleomagnetic data, 10 data sets cannot fulfill the above five criteria and were excluded from our further analyses. For example, paleolatitude estimates of the Late Carboniferous-Early Permian (23.5 ± 2.2°S, X. F. Yang et al., 2017), the Early-Middle Permian (11.2 ± 12.5°S, Cheng et al., 2012), the Late Triassic (30.4 ± 4.0°N, C. Y. Song et al., 2012; 31.8 ± 5.6°N, Zhou et al., 2019) fail Criterion 1 due to their large age uncertainties (~25, 20 and 18 Myr, respectively); Paleolatitude of the Early Permian at 5.1 ± 7.2°S by X. Z. Li et al. (1999) fails Criteria 2–5; Paleolatitudes of the Early-Middle Permian (11.2 ± 12.5°S, Cheng et al., 2012), Middle Permian (16.5 ± 14.6°S, Cheng et al., 2013) and the Late Permian (12.9 ± 9.3°S, Cheng et al., 2012; 5.5 ± 9.3°S, Cheng et al., 2013) fail Criterion two because of the small number of sampling sites (both only have five sites); The Early Triassic paleolatitude of 4.8±6.7°N by Cheng et al. (2012) fails Criterion 2, 4; and the Late Triassic latitude of 31.3 ± 8.8°N by X. Z. Li et al. (1999) fails all Criteria 1–5 (see details of the data evaluation in the Text S6 and Table S3).

The remaining seven paleopoles seem to be high quality results that fulfill all the five criteria and with quality factor (Van der Voo, 1990) $Q \geq 6$ (K. N. Huang et al., 1992; Ma et al., 2019; P. P. Song et al., 2015, 2017, 2020; Zhou et al., 2019; this study) (Table S3). However, after detailed analyses of the original data, the Late

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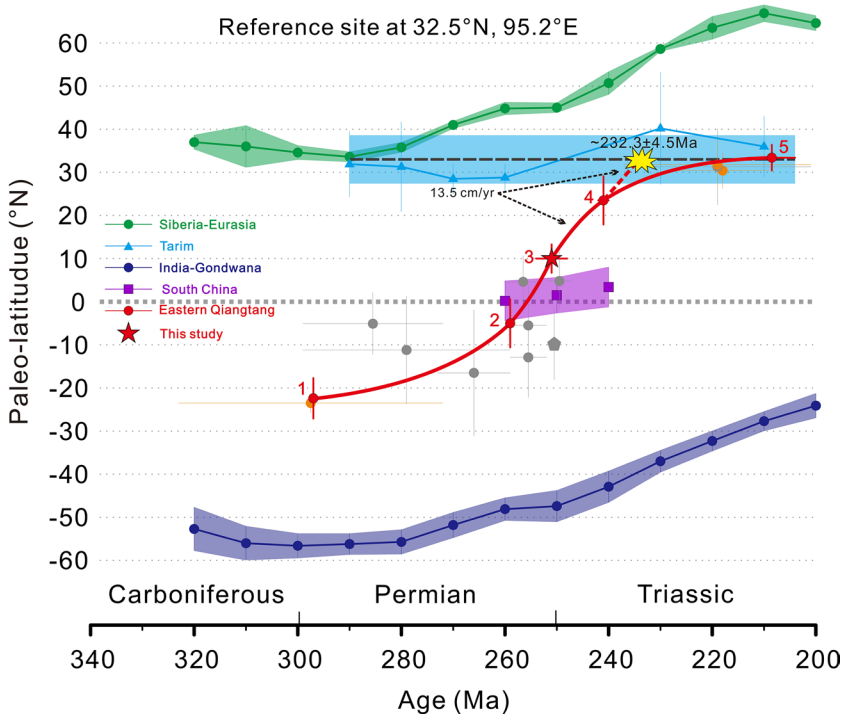


Figure 3. Paleolatitude evolution of the Eastern Qiangtang Terrane (EQT), Tarim, Eurasia and Gondwana during the Late Paleozoic to the Early Mesozoic. Paleolatitudes of Siberia-Eurasia and India-Gondwana are from Torsvik et al. (2012), those of Tarim are from B. C. Huang et al. (2018). The paleomagnetic data of the EQT can be seen in Table S3 (1, P. P. Song et al., 2017; 2, Ma et al., 2019; 3, This study; 4, P. P. Song et al., 2020; 5, P. P. Song et al., 2015), where Tarim, Eurasia and Gondwana are in Table S4. The solid gray pentagon represents the paleomagnetic data from Zhou et al. (2019). The orange filled circles represent the paleomagnetic data less qualified but seems still reliable from C. Y. Song et al. (2012), X. F. Yang et al. (2017), and Zhou et al. (2019).

Permian paleomagnetic result of K. N. Huang et al. (1992) and the Early Triassic result of Zhou et al. (2019) were excluded from our further analyses. For the Late Permian paleolatitude (4.6 ± 6.1°N, $Q = 6$) of K. N. Huang et al. (1992), they assigned the studied sandstone Tuoba Fm. to the Longtanian (Wuchiapingian, ~254–259 Ma), based on the fusulinid *Codonofusiella* in the section, which was later indicated to be older than 268 Ma, based on a recent Rb-Sr data of ~268.4 Ma for the overlying volcanic rock dominated Xiayacun Fm. (TARGS, 2007) (see details in the Text S6). Thus, the age of the paleomagnetic results is controversial and fail Criterion 1. Moreover, even if the assigned Longtanian age were correct, the obtained paleolatitude of 4.6 ± 6.1°N of K. N. Huang et al. (1992) differs from a more robust contemporaneous result from volcanic rocks of 5.0 ± 5.6°S at ~259 Ma (Ma et al., 2019). For the Early Triassic paleomagnetic direction of Zhou et al. (2019), there is an obvious mistake in calculating the mean directions, resulting in a suspicious paleomagnetic pole (invalid even after recalculation), and hence the two results are excluded from our further analysis (see details in the Text S6). Finally, there are only five high quality poles for the Permian-Triassic EQT (Ma et al., 2019; P. P. Song et al., 2015, 2017, 2020; This study), four of which are from the central EQT (e.g., Tuotuohe [P. P. Song et al., 2015, 2017] and Yanshiping [Ma et al., 2019; P. P. Song et al., 2020]), and this study is from the eastern part of the EQT. These high-quality paleomagnetic latitude estimates indicate that the EQT was located at 22.4 ± 4.7°S at ~297 Ma (P. P. Song et al., 2017) and drifted northward to 5.0 ± 5.6°S at ~259 Ma (Ma et al., 2019), moving to 10.0 ± 3.3°N at ~251 Ma (this study), arriving 23.5 ± 5.6°N at ~240 Ma (P. P. Song et al., 2020), and ultimately reaching 33.4 ± 3.0°N during ~213–204 Ma (P. P. Song et al., 2015) (in Zaduo reference). The trajectory indicates that the EQT drifted slowly northward at a rate of ~5.0 cm/yr during ~297–259 Ma, then drifted rapidly northward at a velocity of ~20.6 ± 8.9 cm/yr during ~259–251 Ma and ~13.5 ± 6.5 cm/yr during 251–240 Ma (Figure 3). The latter two estimated velocities are similar to that of the Himalayan Terrane (~15.4 cm/yr) prior to its collision with the Lhasa Terrane (T. S. Yang et al., 2015) and the Indian Plate (15–20 cm/yr) during the ~70–55 Ma time interval (Besse & Courtillot, 2002; Van Hinsbergen et al., 2011; Yi et al., 2011). The acceleration of the drift rate after ca. 259 Ma for

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the EQT was synchronous to the eruption of the Emeishan large igneous province in the SCB during the Late Permian (~260 Ma, Guo et al., 2004). We suspect that the acceleration was likely caused by the eruption due to the pushing away mechanism, as the EQT was located on the west of the SCB, based on the similar bimodal volcanic rocks (Liu et., 2016; J. Wang et al., 2018), paleolatitudes (Ma et al., 2019), and fauna and flora fossils (Metcalfe, 2013; G. C. Zhao et al., 2018) between the EQT and the SCB.

Regardless of inclination bias in sedimentary rocks, among the excluded relatively high quality paleomagnetic results (only fail Criterion 1 with large age uncertainties), the Late Carboniferous-Early Permian (~23.5 ± 2.2°S, X. F. Yang et al., 2017) and the Late Triassic (~30.4 ± 4.0°N, C. Y. Song et al., 2012; ~31.8 ± 5.6°N, Zhou et al., 2019) ones more or less fall along the northward drift path (Figure 3), likely implying that their assigned ages may be reliable. Thus, the three paleomagnetic data sets (C. Y. Song et al., 2012; X. F. Yang et al., 2017; Zhou et al., 2019) are more likely high quality and can be directly used in future related analyses. In addition, for the paleomagnetic result of K. N. Huang et al. (1992), putting their paleolatitude of 4.6 ± 6.1°N on our best fit line will yield a rough mean age of 253 Ma, which is younger than the proposed ages of both K. N. Huang et al. (1992) and TARGS (2007); Further numerical age studies are necessary.

5.2. Implication for the Closure of the Paleo-Tethys Ocean

It is widely accepted that the paleolatitudinal overlap of the EQT with the Tarim Block indicates the collision of the two and the closure of the Paleo-Tethys Ocean (Ma et al., 2019; Song et al, 2015, 2017, 2020; Yan et al., 2016, 2019; Zhou et al., 2019). For the Tarim Block, we adopted B. C. Huang et al. (2018)'s poles and converted their paleolatitudes into our studied location (32.5°N, 95.2°E; Figure 3 and Table S4). There was insignificant paleolatitudinal variation throughout the Permo-Triassic for the Tarim Block, with a mean paleolatitude of ~33.0° ± 5.6°N (Figure 3).

Based on the paleo-positions of the Tarim Block and the drift history of the EQT, we now can evaluate the relative evolution of the Paleo-Tethys Ocean during the Permo-Triassic. (1) The EQT drifted away from the Gondwana supercontinent from 22.4 ± 4.7°S at ~297 Ma (P. P. Song et al., 2017), indicating a wide Jinsha Ocean (eastern Paleo-Tethys Ocean) of ~6,100 ± 560 km (~55.4 ± 5.1° latitudinal differences) between the Tarim Block and the EQT; (2) The EQT reached a paleolatitude of 10.0 ± 3.3°N at ~251 Ma in Gongjue area, resulting in a ~2,500 ± 470 km (~23.0 ± 4.3° latitudinal differences) wide Paleo-Jinsha Ocean at the P/T boundary; (3) The EQT drifted to a paleolatitude of 23.5 ± 5.6°N at ~240 Ma (P. P. Song et al., 2020), implying a ~1,040 ± 610 km wide Paleo-Jinsha Ocean at this time; (4) the EQT reached a paleolatitude of ~33.0° ± 5.6°N, and the EQT overlapped with the Tarim Block at ~209 Ma (P. P. Song et al., 2015). Therefore, we hypothesize that the collision of the EQT with the Tarim Block must have occurred sometime between ~240 and 209 Ma. However, there are no reliable paleomagnetic results in the EQT for this key time period that can be used to estimate the closure age of the Paleo-Tethys Ocean (Figure 3).

Extrapolating a constant northward velocity to estimate a continental collision time has been widely used in tectonic applications of paleomagnetism (e.g., Ma et al., 2014; 2018; 2019; Lippert et al., 2014; T. S. Yang et al., 2015). Based on the age (251–240 Ma) and latitudinal (10°N–23.5°N) differences, a mean velocity of ~13.5 cm/yr was obtained for the EQT. By extrapolating this velocity, it would take ~7.7 ± 4.5 Ma for the EQT to collide with the Tarim Block at ~33.0 ± 5.6°N after 240 Ma. Thus, this velocity estimate would provide a lowermost age of ~232.3 ± 4.5 Ma for the closure age of the Paleo-Tethys Ocean in the region (Figure 3). However, if we assign a best fit line for the paleolatitudes of the EQT at ~297 Ma, ~259 Ma, ~251 Ma, ~240 Ma, and 209 Ma, the collision of the EQT with the Tarim Block would have occurred at ~220 Ma. Therefore, we hypothesize that the collision of the EQT with the Tarim Block and the closure of the Paleo-Tethys Ocean have probably occurred between ~232 and ~220 Ma (Figure 3).

To better understand the collision process of the EQT with the Tarim Block/NCB, some other lines of evidence are considered (Table S5). For example, a recent integrated paleomagnetic study in Yanshiping, Central Qiangtang reveal an apparent change in rotational senses from counterclockwise to clockwise of the Qiangtang terrane during 230–225 Ma, attributed to the Qiangtang - Songpan-Ganzi collision (M. D. Yan et al., 2016); Based on a Late Triassic (Carnian Stage) paleomagnetic result from Yunnan, the EQT-Indo-

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china Terranes were located at ~26°N, a latitude that overlapped with the Tarim Block/NCB, indicating the northern branch of the Paleo-Tethys Ocean was north of the EQT and closed no later than the early Late Triassic (235–230 Ma; Y. G. Yan et al., 2019); A provenance analysis of the light and heavy minerals of Late Triassic sandstone samples from the Western Songpan-Ganzi terrane shows that collision related turbidite deposits commenced at ~230 Ma in this area (K. J. Zhang et al., 2008); Geochronologic and geochemical analysis of intrusive rocks cutting the turbidite deposits in the Songpan-Ganzi terrane indicate that deformation and thickening of the Middle-Late Triassic gravity flow deposits occurred during the Late Triassic (L. Y. Zhang et al., 2014), suggesting an age of collision of Qiangtang Terrane and Kunlun arc of Late Triassic; Collision related unconformities are widespread between the Middle and Upper Triassic strata, such as in the southeastern (e.g., Zaduo, Zhiduo, and Markam; QGS, 2005; TARGS, 2007) and northern margins of the EQT (BGMRXAR, 1993), implying collision between the EQT and the north continents and the closure of the Paleo-Jinsha Ocean occurred by the early Late Triassic; The Upper Triassic molasse deposits are exposed along the northern margins of the EQT (BGMRXAR, 1993), indicating that the syn-collisional event occurred in the Late Triassic; The emplacement age of the hosting rock (i.e., adamellite) of 226 Ma in Gongjue area, indicating that the eclogites had undergone fast exhumation before 226 Ma, and the continent-continent collision of the Zhongza and Qamdo terranes was in the Late Triassic (Tang et al., 2020). Moreover, reliable geologic, paleomagnetic, and paleontologic evidence suggest the closure of the Mianlue Ocean between the southern (e.g., SCB, Indochina, Sibumasu, WQT-Lhasa, and EQT) and the northern (e.g., NCB and North Qinling, Alex, Central Qilian, Tarim, and Qaidam terranes) continents and the formation of the E-W-trending Central China Orogenic System occurred before ~220 Ma (G. C. Zhao et al., 2018). In addition, the Heri granodioritic rocks in the West Qinling area were emplaced in the Late Triassic (235–223 Ma), indicating the closure of the Paleo-A'nimaque-Mianlue Ocean occurred at 235–223 Ma (Xing et al., 2020). Overall, based on the lines of evidence above, we hypothesize that the collision of the EQT with the Tarim Block and the closure of the Paleo-Tethys Ocean probably occurred at ~232–220 Ma, most likely at about 230 Ma (Figure S6).

6. Conclusions

Zircon U-Pb dating of three samples of volcanic rocks from the top, middle and bottom of the Gadikao Fm. provide high quality age control of ~254–248 Ma for this previously assumed Carboniferous or Permian formation. Rock magnetic, field tests and paleosecular variation analyses suggest the presence of a primary, Permo-Triassic boundary magnetization in the samples. Paleomagnetic analysis yield a high quality (29 sites 257 samples) robust paleomagnetic pole of 59.7°N and 228.2°E ($A_{95} = 3.3°$) at ~251 Ma, corresponding to a paleolatitude of 10.0 ± 3.3°N (reference point at 32.5°N and 95.2°E). Integrated with other reliable paleomagnetic data and other lines of geologic evidence, we hypothesize that the closure of the Paleo-Tethys Ocean in the Qamdo region were best estimated at ~232–220 Ma, most likely at about 230 Ma.

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Data Availability Statement

Our data are available from Tables S1–S6 in the supporting information and the Mendeley Data (http://dx.doi.org/10.17632/ngn7s2hmxb.1).

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